

MD. AJIM HOSSAIN

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EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

B.Sc. in Chemical Engineering

Nov 2022 – June 2027(tentative)

- CGPA: 3.70/4.00 (6 semesters completed)
- Relevant coursework: Sustainable Process Engineering, Biochemical Engineering, Process Design, Process Safety, Process Control, Heat Transfer, Mass Transfer, Reaction Engineering, Materials and Energy Balance

RESEARCH INTEREST

- Sustainable Process Systems & Circular Bioeconomy
- Life Cycle Assessment & Sustainable Biorefineries
- CO₂ Capture & Low-Carbon Technologies
- AI/ML for Chemical Engineering

RESEARCH EXPERIENCE

Evaluation of Interchangeability and Re-usability of LCA Modules in a Biorefinery Process Repository

Undergraduate thesis

Jun 2026 – Present

Supervisor: Dr. Kawnish Kirtania

- Developing a generalized, matrix-based Life Cycle Assessment (LCA) methodology to evaluate the interchangeability and reusability of modular LCA components within a P-graph-based biorefinery process repository
- Using citric acid production from beet molasses and corn as a case study; integrating OpenLCA linked to the ecoinvent background database with a Python-based framework for automated result retrieval and cross-module comparison

Cellulose Oxalate Sorbent for Antibiotic Removal from Wastewater

Feb 2024 – Mar 2025

Undergraduate Researcher

- Synthesized a cellulose oxalate sorbent from waste newspaper through chemical modification, using vacuum filtration for sample processing
- Characterized the sorbent using FTIR and UV-Vis spectroscopy to evaluate its structural and functional suitability for removing antibiotic residues from wastewater

PUBLICATIONS & PRESENTATIONS

A Two-Stage Machine Learning-based Framework for Generating and Validating Novel Ionic Liquids for CO₂ Capture

Accepted (Oral), the 8th International Conference on Chemical Engineering (ICChE 2026)

Dec 17–19, 2026

Department of Chemical Engineering, BUET, Dhaka-1000, Bangladesh — Lead Author

- Developed a ChemBERTa-XGBoost ensemble that predicted CO₂ solubility across 10,116 experimental records spanning 124 ionic liquids, achieving test $R^2 = 0.92$ (RMSE = 0.064) under a stricter, ionic-liquid-grouped train/test split comparable to independently reproduced GIN ($R^2 = 0.96$) and GAT ($R^2 = 0.93$) baselines
- Built a quantitative k-nearest-neighbor applicability domain to flag unreliable extrapolations, then screened a combinatorial library of 2,604 ionic liquids (2,480 novel) generated from 37 systematically designed cations, ranking candidates by mass-based CO₂ capacity (kg CO₂/kg IL)
- Verified the top-ranked novel candidates' physical stability via DFT geometry optimization and vibrational-frequency analysis in ORCA (MPI-parallelized), confirming each as a genuine energy minimum and validating the full discovery pipeline end-to-end

TECHNICAL SKILLS

- Programming: Python, MATLAB
- Machine Learning & Cheminformatics: RDKit, ChemBERTa, XGBoost, Graph Neural Networks (GIN, GAT)
- Data Analysis Tools: SPSS, Origin
- Specialized Software: OpenLCA, DWSIM, Aspen HYSYS

- Research Instrumentation: UV Spectrometer, FTIR, GC
- Office Tools: Microsoft Excel, Word, PowerPoint

PROJECT EXPERIENCE

ChE 408: Design of a 10MT/year Cephalosporin API Production Plant (ongoing)

- Conducted market analysis, site selection, material, energy, and utility balances, followed by detailed equipment design, techno-economic evaluation, and risk assessment for a 10 MT/year cephalosporin production plant.

ChE 308: Machine Learning Prediction of CO₂ Solubility in Ionic Liquids Using Thermodynamic and Molecular Descriptors

- Curated and cleaned a dataset of ionic liquid CO₂ solubility measurements using RDKit and standard Python libraries
- Selected and trained candidate machine learning models in MATLAB to predict CO₂ solubility from thermodynamic and molecular descriptors

ChE 306: Simulation and Optimization for Refinery Heat Exchanger Networks

- Simulated the crude oil topping section of Eastern Refinery Limited using Aspen HYSYS
- Optimized the heat exchanger network configuration using HYSYS's built-in optimizer to improve energy efficiency

ChE 302: Analysis of a 100 MTPD Linear Alkylbenzene Sulfonic Acid (LABSA) Production Plant

Process Calculation, Economic Evaluation, and Safety Considerations

- Performed detailed process calculations for a 100 MTPD LABSA production plant, including material and energy balances
- Conducted economic evaluation of the plant and assessed key process safety considerations

INTERNSHIP EXPERIENCE

Industrial Internship

6 – 20 June 2026

Eastern Refinery Limited (ERL)

Chattogram, Bangladesh

- Gained hands-on exposure to plant-wide operations and practical exposure to occupational health and safety practices, including emergency and hazard response protocols, and observed professional conduct expected in an industrial plant environment at ERL, Bangladesh's only petroleum refinery
- Observed the crude distillation unit and adsorption unit, and learned how crude oil is separated into fractional products across interconnected process units and subunits

AWARDS AND SCHOLARSHIPS

- Dean's List Award, two academic years (awarded for maintaining a CGPA of 3.75 or higher)
- Merit List Award, five semesters (awarded for ranking in the top 10% of the class each term)
- Technical Scholarship, six semesters (awarded for satisfactory academic results)

LEADERSHIP & TEACHING EXPERIENCE

Regional Chair

Aug 2025 – Present

Southern Asia Regional Subcommittee, AIChE

- Leading 6 regional liaisons to strengthen communication across 50+ AIChE student chapters in the Southern Asia region

Regional Liaison

Aug 2024 – Aug 2025

Southern Asia Regional Subcommittee, AIChE

- Facilitated communication between 14 chapters across the region and developed strategies to enhance regional engagement and collaboration

Organizing Secretary

Aug 2026 – Present

BUET AIChE Student Chapter

- Leading the executives of the Events Management Team, alongside the Chair, in organizing events such as webinars, seminars, and orientations

Teaching Experience as a private tutor

Chemistry, Physics, and Mathematics

- Tutored around 15 students in basic chemistry, physics, and mathematics
- Provided consistent academic mentorship, exam preparation, and doubt-clearing support over the last 4 years